

Seasonal Climate Effects on Influenza–Pneumonia Mortality and Public Health

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ABSTRACT: We study how seasonal climate affects influenza–pneumonia (I-P) mortality using monthly health and climate data over the past 20 years, reduced to mean annual cycle and statistically correlated. Results show that I-P deaths are inversely related to temperature, humidity, and net solar radiation in the United States, South Africa, and Puerto Rico ($r < -0.93$) via transmission and immune system response. The I-P mortality is 3–10 times as high in winter as in summer, with sharp transitions in autumn and spring. Public health management can rely on seasonal climate-induced fluctuations of I-P mortality to promote healthy lifestyle choices and guide efforts to mitigate epidemic impacts.

KEYWORDS: Subtropics; Regression analysis; Statistics; Disease; Seasonal effects

1. Introduction

The world has seen repeated viral epidemics and interventions aimed at preventing community transmission. Viral risks are attributed to host, viral, bacterial, and environmental factors. Host factors include underlying comorbidity and immune dysfunction. Viral factors involve polymerase mutations, host switch/adaptation, and change of viral proteins, which inhibit immune and antiviral responses. Influenza–pneumonia (I-P) can cause mortality in older people at high risk. Worldwide, annual I-P outbreaks contribute 3–5 million cases of severe illness and 300 000–600 000 deaths (World Health Organization 2018). Mortality rates from I-P are greater among the poor and in people over age 75 (Centers for Disease Control and Prevention 2017). Globally, about 0.06% of all deaths are attributed to I-P, amounting to ~ 4.2 million yr^{-1} (Iuliano et al. 2018). Poor hygiene, lack of potable water/electricity, and nutrient-poor diets linked to poverty can contribute to higher I-P mortality.

Six global pandemics have emerged since 1889 and led to health impacts in the following winter (Fox et al. 2017). Epidemics expand their geographic reach as population-wide immunity falls and I-P transmission rises during the winter months (Fox et al. 2017). Improved global epidemic surveillance show that Southeast Asia is a common source of I-P viruses (Bedford et al. 2015) due to dense population and livestock. Disease severity increases in pandemic viruses that derive from zoonotic sources (Kash and Taubenberger 2015).

Climate affects lifestyle and therefore plays a role in immune system functioning (Hahtela et al. 2019) via pathogen recognition and receptor mitigation. Vulnerability to infectious disease involves age, choices regarding physical activity and diet, and comorbidities. Living in proximity to nature, consuming fresh fruits and vegetables, and avoiding sugar, drugs, and obesity tends to strengthen immunity

(Childs et al. 2019). Human defenses against disease are better in summer when outdoor exposure to sunlight and an accelerated cardiovascular circulation improves the uptake of nutrients, minerals, and vitamins (Cannell et al. 2006; Urashima et al. 2010; Schwalfenberg 2011; Fares 2013). The immune system atrophies as we reach life-span and during winter season. Mortality rises exponentially with age (Verity et al. 2020; Fig. 1a). There is an increasing spread over a life-span that is attributed to individual “resilience” and nonclimatic factors such as exercise, diet, and waist–height ratio. Health experts can help reduce the risk of mortality from infectious diseases by educating the public on how transmission and immunity are compromised in winter. The inevitability of I-P virus mutations means that vaccines offer temporary solutions.

Precipitation and temperature, population density, travel patterns, and time spent indoors drive I-P mortality risk (Amendolara 2019). Seasonal I-P transmission emerges many months after the initial spread of mutated viruses that escape immunity, and surges according to latitude: during winter in temperate zones and during rainy spells in the tropics (Dowell and Ho 2004; Deyle et al. 2016). In midlatitudes, lower temperatures tend to confine activity indoors, causing closer contact under reduced ventilation that increases I-P transmission. Colder temperatures tend to increase the longevity of viruses and host viral shedding. Low ultraviolet (UV) light contributes to seasonal influenza via reduced production of melatonin and vitamin D, the lack of which causes immune deficiencies (Nelson and Drazen 1999; Amendolara 2019).

A distinct attribute of coastal zones is sea spray aerosols ($>1 \mu\text{m}$) generated by wind waves. Marine air contains a variety of alkaline mineral salts and is low in pollutants such as ozone because of halogen reduction (Saiz-Lopez and Fernandez 2016). Sustained inhalation loosens mucus and increases oxygen levels that stimulate immunity in the skin and lungs. Humid salty air inactivates aerosolized viruses (De Jong et al. 1973; Yang and Marr 2012). Land–sea temperature differences drive

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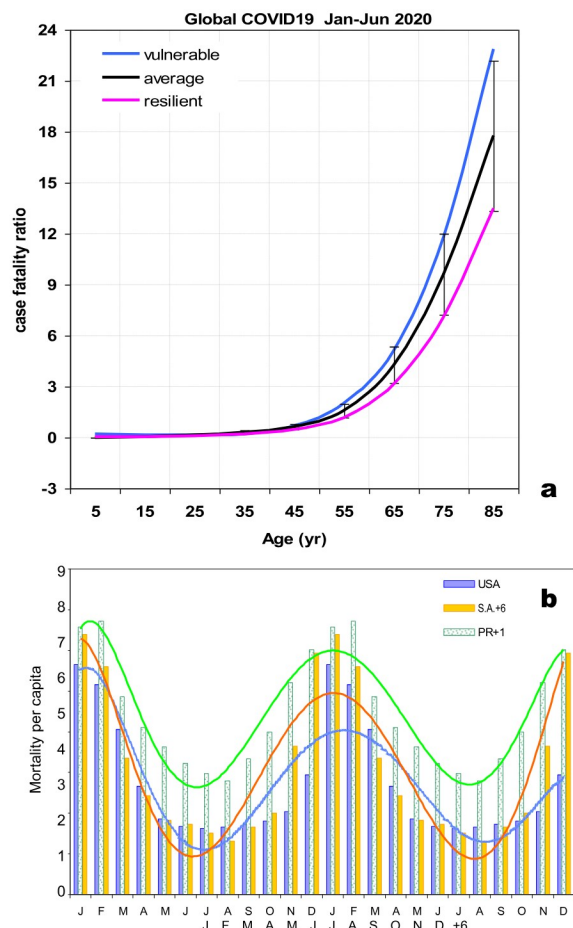


FIG. 1. (a) Case fatality ratio (%) by age class for coronavirus disease 2019 (COVID-19) I-P virus based on data from Verity et al. (2020): vulnerable = less active, poor diet, and waist–height ratio > 0.6 ; resilient = more active, better diet, and waist–height ratio < 0.45 . (b) Mean annual cycle of I-P mortality normalized by population ($\times 10^{-5}$) fitted with a polynomial function. South Africa uses a lower scale for Southern Hemisphere winter. Note that annual cycle graphs are repeated $\times 2$ to place the winter peak in the middle, here and elsewhere.

marine air inland during summer and offshore during winter. In contrast, midlatitude interior locations have more airborne particulates and vulnerability to respiratory infections in winter (Rosano et al. 2019).

Lifestyle factors including diet and exercise alter the risk of complications from infectious diseases (Fares 2013). Eating a healthy diet of proteins, vegetables, and fruits ensure a healthy immune system. Obesity is present in $\sim 50\%$ of population in industrialized countries and is a growing problem (Luzi and Radaelli 2020). Obesity increases infection risk and give rise to virulent strains. Obese people often develop an inflammatory response and lung lesions when contracting I-P. Lack of exercise and insulin resistance are characteristic of obese people and impair resistance to microbial infections (Luzi and Radaelli 2020). They tend to shed the I-P virus for

longer periods than thin people and can thus infect more, prolonging an epidemic outbreak. There is evidence that low sugar and carbohydrates, high fiber, and protein diets rich in certain fats help to protect against I-P symptoms throughout the world (Goldberg et al. 2019).

Frequent outdoor aerobic exercise enhances I-P immunity (Campbell and Turner 2018; Song et al. 2020) and reduces I-P infection risk. There is epidemiological evidence that long-term, daily, structured exercise limits chronic and communicable diseases such as viral and bacterial infections in the elderly (Campbell and Turner 2018; Song et al. 2020) and enhances immune mitigation of I-P. Exercise also limits atrophy of the immune system during aging and enhances the T-cell pool across the life-span (Campbell and Turner 2018). Thus, by promoting exercise in the elderly population who carry the greatest health risk, the entire community benefits.

Viral transmission occurs by airborne droplets, contact and fomites (Tellier et al. 2019). All I-P viruses (including corona type) have envelope proteins, which are inactivated by environmental factors, such as humidity, making them infective (Centers for Disease Control and Prevention 2017; Wang et al. 2021). They undergo wet deposition, settling on surfaces that remain infectious under cool or artificially dry conditions (indoors), but deteriorate outdoors at higher temperature and humidity (Casella et al. 2020). Ambient water vapor pressure is controlled by sources of moisture and air temperature, and typically exceeds 20 hPa outdoors during summer and in tropical marine air masses. I-P mortality increases in temperate zones with lower vapor pressure in the winter hemisphere. The slow spread of harmful viruses to lower latitudes and the summer hemisphere is attributed to distance from origin, and to environmental controls on transmission and immunity (Lowen et al. 2007). Understanding the seasonal environmental factors that affect I-P mortality could assist in mitigating pathogenic impacts.

In the following work, monthly mean I-P mortality and climate descriptors show marked transitions that could be used, in conjunction with improved lifestyle and appropriate health education, in managing epidemics. Zoonotic corona viruses such as severe acute respiratory syndrome coronavirus 1 (SARS-CoV-1) and 2 (SARS-CoV-2) that infect the human population, show similar drivers and seasonal consequences after initial emergence and penetration into the community. Summertime lockdowns have led to questioning whether such measures could rather be timed to cover winter upturns.

Here, the eastern United States and Caribbean island of Puerto Rico were selected as Northern Hemisphere locations, and South Africa was selected as a Southern Hemisphere location. Our main objective is to analyze the correlation between the mean annual cycle of climate and I-P mortality across geographically diverse settings. We then offer recommendations that align transmission controls and immune-boosting activities with our findings.

Our scope of work is limited to national statistics and excludes a spatial analysis of I-P mortality.

2. Methods

a. Health data

Mortality ascribed to influenza often includes pneumonia associated with corona viruses ([appendix Fig. A1](#)). Where they are listed separately, monthly deaths attributed to I-P were summed from national health registers over the past 20 years within geographic country boundaries, which for the United States refers to the mainland (see [Centers for Disease Control and Prevention, 2020](#)). For South Africa, mortality statistics attributed to I-P are obtained via ([Statistics South Africa 2020](#); account required for access) ([appendix Fig. A2](#)) as in earlier scientific studies ([Dangor et al. 2014](#); [Kyeyagali et al. 2014](#); [Cohen et al. 2010, 2018](#)).

b. Climate descriptors

Climate data over the past 20 years derive from numerical reanalysis via ERA5 ([Hersbach et al. 2020](#)) and MERRA-2 ([Molod et al. 2015](#); [Randles et al. 2017](#)) and describe surface net ultraviolet solar radiation, surface air temperature, water vapor pressure, surface evaporation (or moisture flux), rainfall, surface winds, carbon monoxide (pollution), and (sea) salt spray via websites ([IRI 2020](#); [KNMI 2020](#); [NASA 2020](#)). For Puerto Rico, the area average for climate parameters covers the island 18°–18.5°N, 67°–66°W; for the United States, climate data were averaged over “west” (30°–45°N, 130°–100°W) and “east” (25°–45°N, 100°–70°W) sectors, excluding sea. Similarly, climate data for South Africa were averaged over “west” (34°–30°S, 17°–26°E) and “east” (34°–24°S, 26°–33°E) sectors, excluding sea. In addition to climate data, satellite color fraction (NDVI; [Tucker et al. 2005](#)) is used to represent vegetation transpiration, which lags climate, as do health impacts. ERA5 is employed for climate parameters except satellite vegetation, MERRA-2 salt spray and carbon monoxide.

c. Statistics

All data are reduced to their mean annual cycle for statistical analysis, for example, ~20 of each month; consistent with [Caini et al. \(2017\)](#). The annual cycle of I-P mortality is correlated with climate descriptors per country. East/west sectors are used for large continents and 0- and 1-month lag correlations were calculated. Statistical confidence is reached with a Pearson product cross-correlation value < -0.90 with 12 degrees of freedom. The correlation determines how well the climate annual cycle fits the shape and phase of the I-P mortality data. We use mortality to reflect more lethal viruses, instead of hospital case admissions driven by seasonal flu, the common cold, or asthma ([Lewis et al. 2020](#)). Thus, we establish the seasonal climate sensitivity of stronger corona virus strains causing death, for example, SARS-CoV-1 or Middle East respiratory syndrome coronavirus (MERS-CoV-2) ([Lin et al. 2006](#); [Bloom-Feshbach et al. 2013](#); [He et al. 2015](#); [Nassar et al. 2018](#)), to offer insights on future epidemics.

Graphical presentations show two annual cycles ([Fig. 1b](#)) to place the winter peak in the middle; with climate descriptors inverted. In addition to temporal statistics, a mapping is done

for mean climate variables associated with I-P mortality, focusing on the winter season and the geographic distribution of surface wind and temperature; and MERRA-2 satellite-assimilated sea salt aerosols and carbon monoxide concentrations that affect respiratory function.

d. Focus

Based on the above literature review, we understand that the emergence of new I-P viruses and their initial impact (e.g., first wave of infection) are unrelated to climate. It is only after penetration into the community that the seasonality of transmission and immunity takes hold. Furthermore, we do not expect to find (or analyze for) year-to-year variations in I-P mortality, as corona viruses often emerge via zoonotic mutation from sources far away from human receptors (via international air travel), thus under different climatic conditions.

3. Results

a. Puerto Rico

Caribbean islands are surrounded by salty marine air carried by subtropical trade winds ([Figs. 2a–c](#)). Carbon monoxide pollution hot spots are evident on 18°N latitude at Santo Domingo (70°W), Port au Prince (72°W), and Kingston (77°W). These cities are located on leeward coasts, unlike San Juan (66°W). Only the elevated interior of Hispaniola has reduced salt aerosols; elsewhere, trade wind waves induce high values that wrap around the islands. A scatterplot ([Fig. 2a](#)) compares the mean annual cycle of water vapor pressure and Puerto Rico I-P mortality. Mortality declines threefold from winter to summer with inhibited virus transmission ([Hirve et al. 2016](#); [Martinez 2018](#)). The average rate of I-P mortality is ~1500 deaths per year.

Cross correlations of Puerto Rico I-P mortality and the mean annual climate are listed in [Table 1](#). Seasonal changes in rainfall are insignificant. Negative correlations are strong for vapor pressure, temperature, and moisture flux ([Fig. 2d](#)) with mortality lagged one month (+1). Seasonal I-P deaths decline from ~230 to ~70 per month in summer. For net UV radiation, the correlations are higher in the same month ([Fig. 2e](#)) indicating that outdoor exposure to sunshine has immediate health benefits. It can be noted that I-P infections in Puerto Rico often derive from U.S. visitors ([Paz-Bailey et al. 2020](#)), and that a marine climate tends to lag terrestrial climate by ~1 month.

b. United States

Health data represent U.S. national figures, but different parts of the country are so interconnected that the normalized annual cycle of I-P mortality in east and west sectors varies by less than 10% in magnitude and phase. The United States experiences ~56 000 deaths per year on average ([appendix Fig. A2](#)). The high point is ~12 000 month⁻¹ in January, and the low point is ~1900 month⁻¹ in July, a sixfold seasonal change. December I-P mortality fluctuates more than other months, and the high point often shifts into February. The U.S. annual cycle of I-P mortality is compared with the

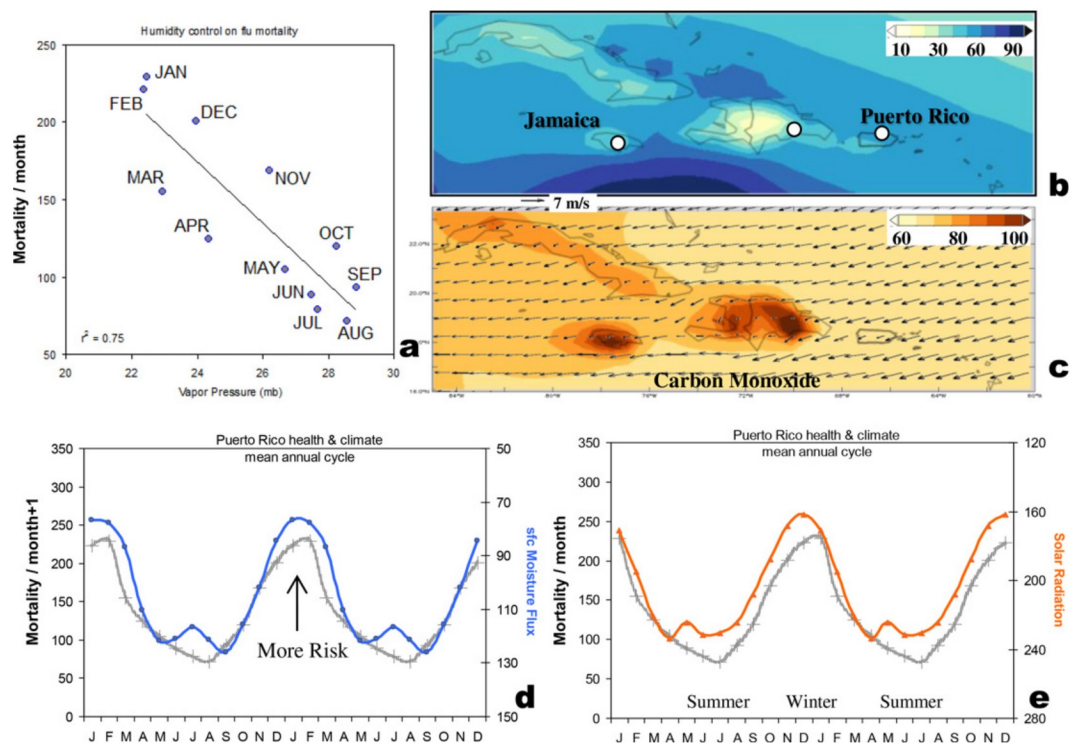


FIG. 2. (a) Scatterplot of monthly mean water vapor pressure and Puerto Rico I-P mortality. Also shown are Caribbean maps of annual average concentration of (b) sea salt aerosols (ppb) and (c) carbon monoxide (ppb) and winds, along with the mean annual cycle $\times 2$ of Puerto Rico I-P mortality with (d) (inverted) lagged evaporation or moisture flux (W m^{-2} ; mortality + 1 month) and (e) (inverted) net solar radiation (W m^{-2}). Major cities are shown by dots in (b).

vegetation index in Fig. 3a and is neatly contained by seasonal changes in both east and west sectors.

The mean annual cycle of net UV radiation, (evaporation) surface moisture flux, surface air temperature, and vegetation

fraction over the United States are significantly correlated with national I-P deaths in the (+1) following month at $r < -0.90$, Table 1. The annual cycle of water vapor pressure over east and west United States (Fig. 3b) exhibits the

TABLE 1. Correlation between climate and national I-P mortality data averaged over the annual cycle; sample size of $N = 12$, and $r \leq -0.90$ being significant (given in boldface type). Here, mortality refers to I-P, + 1 indicates lagged 1 month, E (east) and W (west) refer to climate sector (Puerto Rico has only one sector), netUV = net solar radiation, evap = evaporation (or moisture flux), temp = air temperature, vapor = water vapor pressure, rain = rainfall, and veget = vegetation index. Weak correlations are excluded.

	Puerto Rico		USA		South Africa Mortality
	Mortality	Mortality + 1	Mortality	Mortality + 1	
Mortality + 1	0.82	—	0.78	—	—
NetUV	-0.97	-0.74	—	—	—
E netUV	—	—	-0.66	-0.92	-0.82
W netUV	—	—	-0.69	-0.92	-0.84
Evap	-0.72	-0.95	—	—	—
E evap	—	—	-0.76	-0.90	-0.91
W evap	—	—	-0.71	-0.91	-0.94
Temp	-0.68	-0.93	—	—	—
E temp	—	—	-0.87	-0.92	-0.96
W temp	—	—	-0.82	-0.90	-0.94
Vapor	-0.49	-0.85	—	—	—
E vapor	—	—	-0.82	-0.83	-0.93
W vapor	—	—	-0.77	-0.77	-0.89
E veget	—	—	-0.88	-0.86	-0.81
W veget	—	—	-0.86	-0.92	—
Rain	-0.24	-0.64	—	—	—

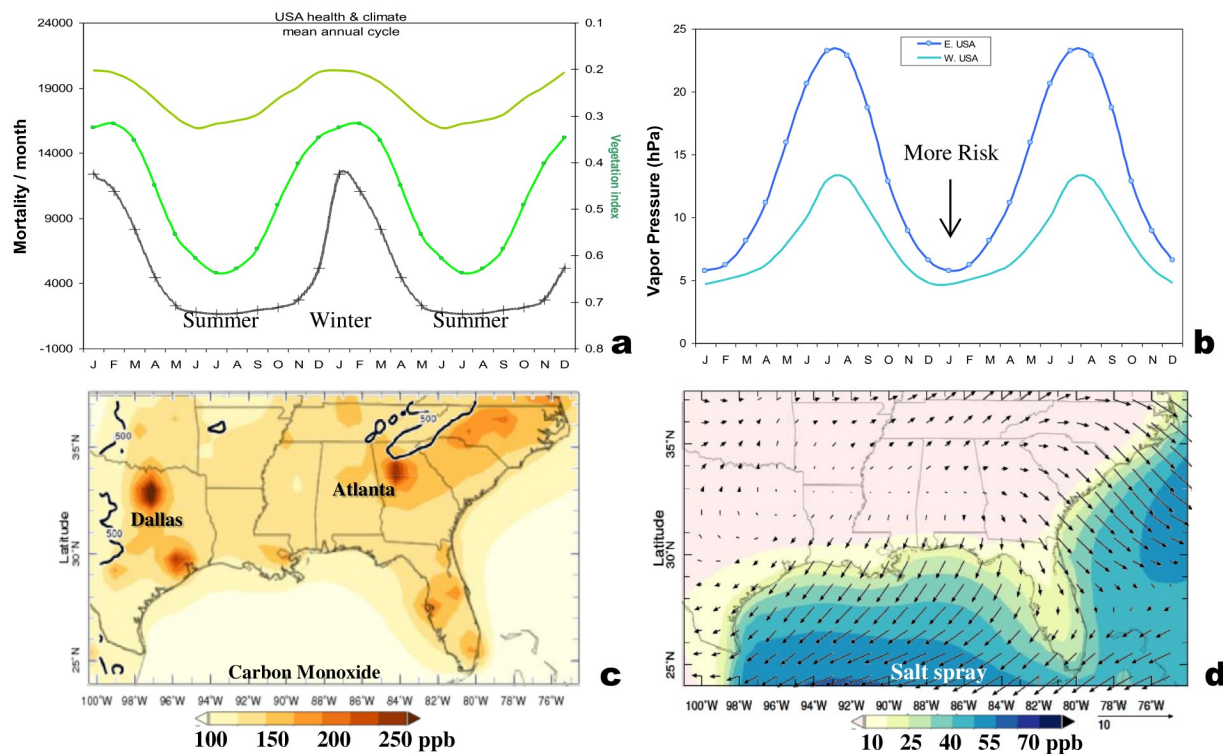


FIG. 3. Mean annual cycle $\times 2$ of (a) U.S. I-P mortality (black line) and (inverted) vegetation index (%) in western (olive line) and eastern (bright-green line) areas, and (b) east/west water vapor pressure. (c) Southeastern U.S. map of winter carbon monoxide concentration (ppb) and urban hot spots, with 500-m elevation contour. (d) Map of winter-season wind (m s^{-1}) and sea salt aerosols (ppb). Spatial maps in (c) and (d) cover the eastern U.S. area and show a variety of conditions that are lumped together in regional averages.

expected Gaussian peak in July–August when the threat of I-P mortality is low.

Figure 3c considers the geography of winter air pollution and reveals carbon monoxide hot spots such as Atlanta and Dallas situated ~ 500 km inland. The Florida Peninsula has similar urban emissions that are dispersed by coastal winds. The map of winter winds and sea salt spray (Fig. 3d) that alters transmission and immunity (Zhang et al. 2006; Yang and Marr 2012; Parks et al. 2018) is affected by cool land temperatures. The warm Gulf Stream around Florida supplies salty air toward southeast-facing coasts as far as Texas. However, sea salt aerosols undergo rapid deposition and seldom penetrate far inland, particularly in winter when offshore winds are prominent.

c. South Africa

A Southern Hemisphere location will offer contrasts in the timing of winter impacts on health. Carbon monoxide pollution is evident (Fig. 4a) over Gauteng (population 14 million), situated ~ 500 km inland at ~ 2000 m elevation. Air pollution could magnify the annual cycle of I-P mortality, particularly in dense and poor populations with limited access to potable water that compromises hygiene. During winter when minimum temperatures drop below freezing, charcoal stoves

provide heat—but ventilation is poor and the indoor air becomes polluted.

There are $\sim 14\,000$ I-P deaths per year in South Africa's population of 55 million, with a greater burden of acute respiratory disease (Cohen et al. 2015). The minimum rate of I-P mortality is January (summer) and there are sharp transitions in May and August; seasonal differences are ~ 10 -fold (Fig. 4b). Annual cycle correlations (Table 1) are simultaneous and negative for vapor pressure ($r = -0.93$) and temperature ($r = -0.96$), which exhibit softer seasonal transitions than I-P mortality. Maps of South Africa winter climate (Figs. 4c,d) reveal a pool of salty air over the warm Agulhas Current, which seldom reaches the coast because of an offshore circulation. Winter temperatures are warmer on the southeastern coast but decline rapidly inland over the Highveld where dry frosty conditions prevail from May to August. Cities on the plateau are thus at greater health risk.

4. Discussion

a. Climate-health outcomes

Given that past I-P mortality involves corona viruses, it is believed that historical inferences are valid. Climate is correlated negatively with health risks, for example, warmer weather and higher vapor pressure limits infectiousness, regardless of

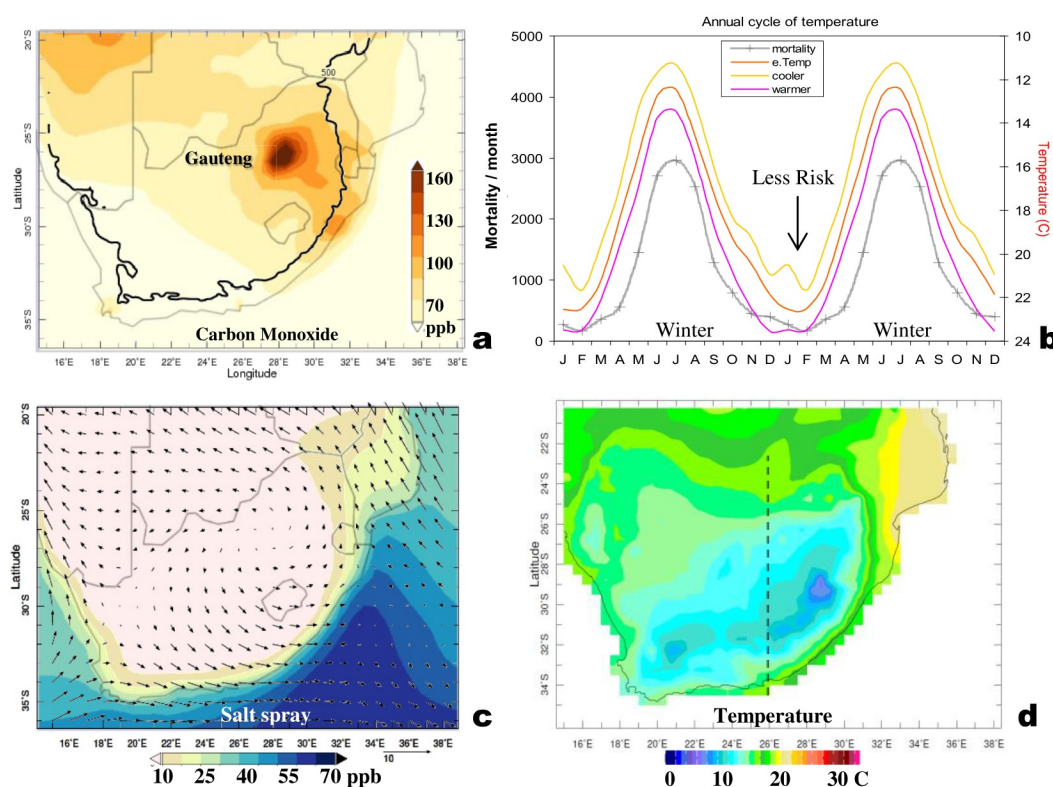


FIG. 4. (a) South Africa average winter-season carbon monoxide (ppb) and urban hot spots with 500-m elevation contour. (b) Annual cycle $\times 2$ of (inverted) surface air temperature mean, minimum, and maximum ($^{\circ}\text{C}$) and I-P mortality. (c) Maps of winter-season wind (m s^{-1}) and sea salt aerosols (ppb), and (d) winter surface air temperatures. Dashed line in (d) separates west and east regions.

geography or health statistics. The annual cycle of I-P mortality correlated at < -0.90 with temperature and humidity in all three countries, albeit with a 1-month delay in the United States and Puerto Rico. Winter weather desiccates the air, traps air pollutants (World Health Organization 2013), forces people indoors, and slows the cardiovascular circulation, so health risks grow (Fares 2013). Cold spells and reduced sunshine in midlatitudes yield a lag effect on mortality that extends into spring season (Stewart et al. 2017; Kalkstein et al. 2018). Along with greater exposure to ambient pollutants such as oxides of nitrogen, sulfur, and carbon, a lack of sunlight reduces vitamin D formation and lowers immunity to respiratory infections.

Deaths are higher in cold seasons when reduced sunshine and increased air pollution combine with psychological depression. Regional differences in climate (in large countries) have little effect on the annual cycle of health.

Puerto Rico, United States, and South Africa had lower mortality in summer by a factor of 3, 6, and 10, respectively. Amplitude can be traced to the marine versus continental nature of climate. In Puerto Rico's subtropical marine air, there was less seasonal change in health risk. The heterogeneous seasonal weather of the United States induced moderate fluctuations in health risk. In contrast, South Africa's dry winter climate (excepting Cape Town) turns health risks on in

May and off in August. The shape of the winter peak in mortality varies slightly (Fig. 1b): Puerto Rico and United States were asymmetric early/late, respectively, while South Africa was near-Gaussian and in phase with solar angle. Heat and sunlight promote outdoor lifestyles that limit viral infections. Salty marine air spills across subtropical islands and over coasts during summer, but seldom reaches inland in winter. Yet climate acts in parallel with hygiene and lifestyle choices: communities with lower income and poor diets have proportionately higher I-P mortality. Community transmission is inhibited outdoors in warm humid air (appendix Fig. A3); most corona virus infections occur indoors under cool dry ventilation (Centers for Disease Control and Prevention 2020; Dietz et al. 2020). Indoor air measurements taken in offices and food stores in Puerto Rico during March 2020 gave water vapor pressure below 13 hPa, despite outdoor values above 26 hPa. Risks could be reduced by setting air-conditioning thermostats higher.

Based on the spatial analysis and inferences on climate (Metcalf et al. 2017), areas of lower health risk include sunny subtropical zones, coastal margins, and the summer hemisphere, in the absence of air conditioning. The sea salt aerosol maps suggest improved health conditions on warm southeast-facing coasts. Net UV radiation tends to rise quickly in spring with direct sun angle and delayed buildup

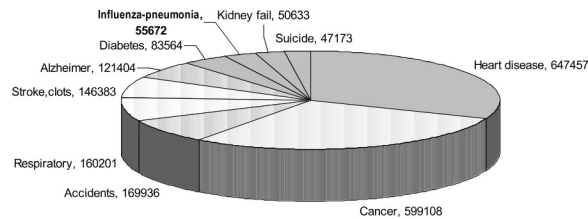


FIG. A1. Background mortality rate as based on annual statistics of U.S. mortality averaged over the past decade. I-P is used here to study climate sensitivity.

of seasonal cloudiness. Epidemics arriving in summer will have less impact and foster population immunity.

Corona viruses derived from mutated zoonotic sources that have infected the human population (Sironi et al. 2020) demonstrate similar transmission/immunity drivers and seasonal consequences (Davis et al. 2012, 2016). The latest outbreak in 2020 puts the global I-P mortality rate above 0.1%, against a background risk from other diseases, cancer, and accidents (appendix Fig. A1) of ~1% or 70 million deaths per year worldwide. Across a variety of geographic settings there is a four month window of health risk, despite epidemic waves that shift the curve. Springtime reductions in mortality can be expected because of the parallel factors of transmission and immunity (Moriyama et al. 2020).

According to the results here, mortality is markedly seasonal over a variety of pathogens. Relying on winter–summer transitions and immune conditioning (see appendix section c) instead of prevention, could lead to an earlier closure of epidemic impacts.

b. Limitations of the study

The study limitations include (i) the use of national level health data, (ii) assumption that mortality is correctly diagnosed, (iii) assumption that the annual cycle of I-P mortality offers historical context for future epidemics, (iv) that climate conditions in east and west sectors of large continents have common influences on national health, and (v) assumption that seasonal climate alters both lifestyle immunity and transmission factors.

5. Conclusions

The mean annual cycle of I-P mortality has been associated with a variety of climate descriptors, to determine historical trends in infectiousness in three geographical locations. I-P deaths follow temperature and humidity in Puerto Rico, a subtropical island ($r < -0.93$). Temperature and vapor pressure are linked with the annual cycle of mortality in the United States and South Africa ($r < -0.91$). Net UV solar radiation gave a simultaneous correlation that suggests immunity acts in parallel with transmission. Ambient winter weather effects on respiratory infectiousness were mapped to show that east coasts may be at lower risk. The summer months exhibit fewer I-P deaths than winter by a factor of 3–10. Significant negative correlations between multiple climate elements

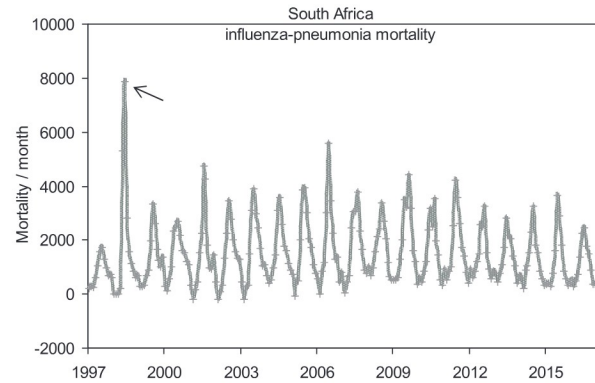


FIG. A2. Raw data, showing an example of the monthly health record used to form the mean annual cycle, as reference for climate sensitivity analysis in South Africa. The arrow points to a prior epidemic.

and I-P mortality provide historical inferences to anticipate seasonal transitions.

Multiple factors influence the spread of I-P in communities and include climatic seasonality, temperature and humidity, sunshine, lifestyle, physical activity, age, diet–obesity, poverty, and living conditions. The importance and relevance of the study emerges from a recognition that climate affects both lifestyle immunity and transmission factors and can be relied on to manage I-P epidemics and avoid economically depressing lockdowns in the summer half year.

Data availability statement. Data are derived from websites of the CDC, StatSA, KNMI Climate Explorer (<https://climexp.knmi.nl/start.cgi>), and IRI Climate Library as listed in the text. A data spreadsheet is available on request.

APPENDIX

Supporting Material

a. Background mortality

Figure A1 shows the U.S. background mortality rate as a pie chart, wherein we see that the I-P category ranks eighth.

b. Raw data

Figure A2 illustrates monthly raw data for South Africa. Note the I-P epidemic in winter 1998. Seasonal peaks of mortality occur as follows: 5 May, 11 June, and 4 July.

c. Some factors in I-P infection

Transmission is one factor: over 90% of all SARS-CoV-2 infections have been traced indoors (food shopping, restaurants, and bars) and could be reduced by social distancing and masks; few infections have been traced outdoors (Morawska and Cao 2020).

Risk is lowered by proximity to a warm coast, a healthy lifestyle, and a good diet (low sugar and carbohydrates, high fiber, and protein). Immunity is boosted by outdoor

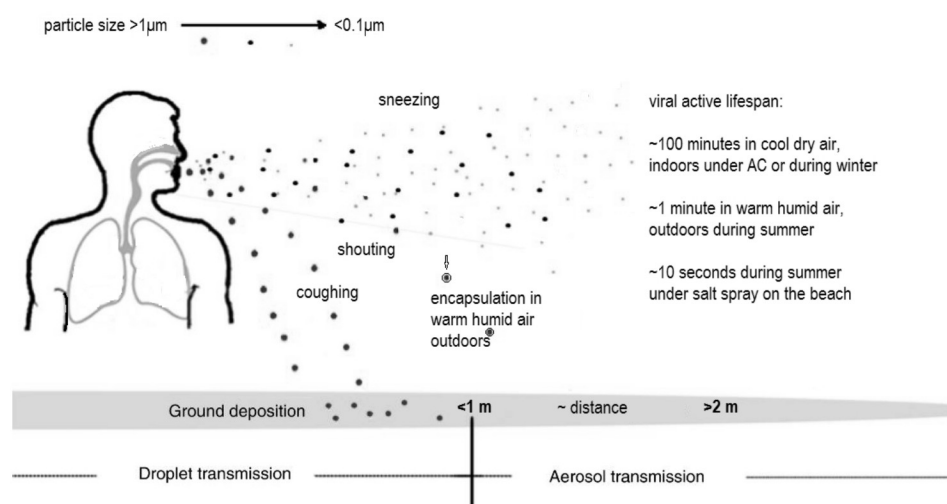


FIG. A3. Schematic diagram adapted from the University of Florida website Explore Research/A question of physics (virus transmission; <https://explore.research.ufl.edu/a-question-of-physics.html>), and the SARS airborne virus lifespan calculator from the U.S. Department of Homeland Security (<https://www.dhs.gov/science-and-technology/sars-airborne-calculator>), giving support to indoor mask wearing.

exercise $>1 \text{ h day}^{-1}$, sunshine and sports, and nature walks (Fares 2013). These strategies may reduce health risks during winter.

d. Schematic diagram

Environmental factors affecting microscale virus transmission are shown in Fig. A3, which also highlights how humidity encapsulates airborne particles.

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